

Problem Set 1

Group 2

2026-01-29

1.) Prove $\lim_{x \rightarrow -1} 2x + 1 = -1$

Determine $\delta > 0$ so that if $|x - (-1)| < \delta$ then $|2x + 1 - (-1)| < \epsilon$.

$$|(2x + 1) - (-1)| < \epsilon$$

$$|2x + 2| < \epsilon$$

$$2|x + 1| < \epsilon$$

$$2|x - (-1)| < \epsilon$$

$$|x - (-1)| < \frac{\epsilon}{2}$$

Given $\epsilon > 0$, Choose $\delta = \frac{\epsilon}{2}$, thus,

$$|x + 1| < \delta$$

$$|x + 1| < \frac{\epsilon}{2}$$

$$2|x + 1| < \epsilon$$

$$|2x + 2| < \epsilon$$

$$|2x + 1 + 1| < \epsilon$$

$$|2x + 1 - (-1)| < \epsilon$$

2.) Determine all the numbers c which satisfy the conclusions of the Mean Value Theorem for the following function and graph using R with the point/s identified. $f(x) = x^3 - 4x^2 - 2x - 5$ on $[-10, 10]$.

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$3c^2 - 8c - 2 = \frac{[(10)^3 - 4(10) - 2(10) - 5] - [(-10)^3 - 4(-10) - 2(-10) - 5]}{10 - (-10)}$$

$$3c^2 - 8c - 2 = \frac{575 - (-1385)}{20}$$

$$3c^2 - 8c - 2 = 98$$

$$3c^2 - 8c - 100 = 0$$

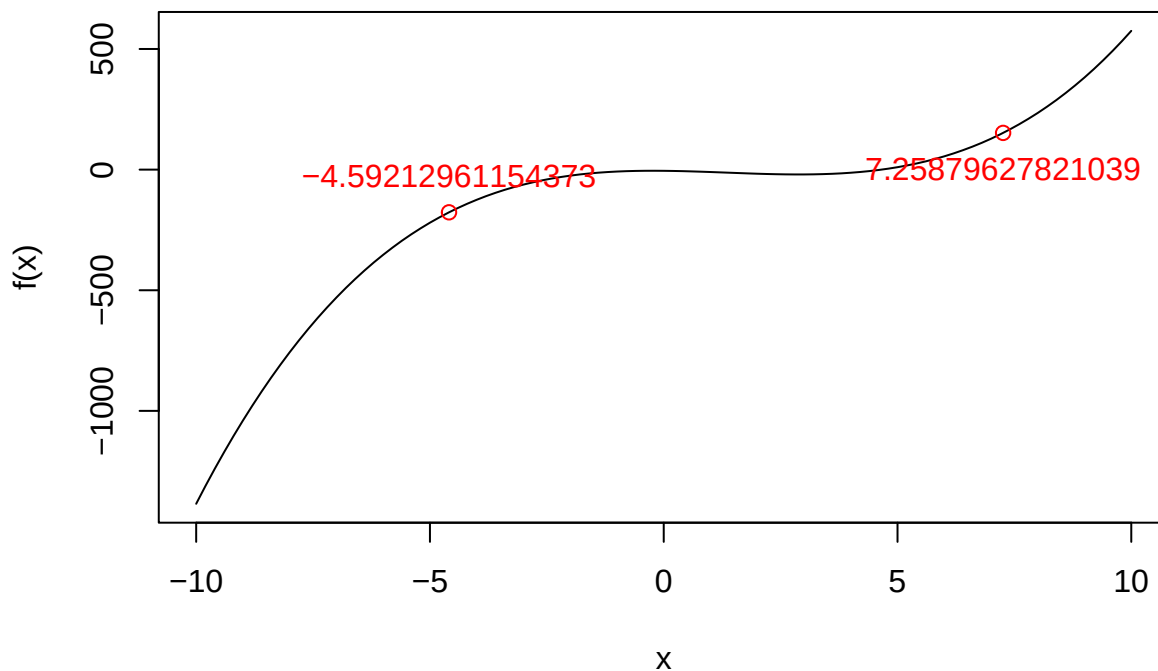
$$c = \frac{-(-8) \pm \sqrt{(-8)^2 - 4(3)(-100)}}{6}$$

$$c = \frac{8 \pm \sqrt{1264}}{6}$$

$$c = \frac{8 \pm 4\sqrt{79}}{6}$$

$$c \approx -4.59$$

$$c \approx 7.25$$



3.) Find the point c that satisfies the mean value theorem for integrals on the interval $[-1, 1]$. The function is $f(x) = 2e^x$

$$\begin{aligned}
\int_{-1}^1 2e^x dx &= (2e^c)(1 - (-1)) \\
2e^x \Big|_{-1}^1 &= (2e^c)2 \\
(2e - 2e^{-1}) &= 4e^c \\
\frac{2(e - e^{-1})}{4} &= e^c \\
\frac{(e - e^{-1})}{2} &= e^c \\
\ln\left(\frac{(e - e^{-1})}{2}\right) &= c \\
0.16143936157 &\approx c
\end{aligned}$$

4.) Consider the function $f(x) = \cos(x)$

a. Find the fourth Taylor polynomial for f at $x = \pi$.

$$\begin{aligned}
f(x) &= \cos(x) \rightarrow f(\pi) = -1 \\
f'(x) &= -\sin(x) \rightarrow f'(\pi) = 0 \\
f''(x) &= -\cos(x) \rightarrow f''(\pi) = 1 \\
f'''(x) &= \sin(x) \rightarrow f'''(\pi) = 0 \\
f^{(4)}(x) &= -\cos(x) \rightarrow f^{(4)}(\pi) = -1
\end{aligned}$$

$$P_4(x) = -1 + \frac{1}{2}(x - \pi)^2 + \frac{1}{24}(x - \pi)^4$$

b. Use the fourth Taylor polynomial to approximate $\cos(0)$.

Substitute $x = 0$:

$$\begin{aligned}
P_4(0) &= -1 + \frac{1}{2}(-\pi)^2 + \frac{1}{24}(-\pi)^4 \\
P_4(0) &= -1 + \frac{(-\pi)^2}{2} + \frac{(-\pi)^4}{24}
\end{aligned}$$

Numerically,

$$\begin{aligned}
P_4(0) &\approx -0.124 \\
\cos(0) &\approx -0.124
\end{aligned}$$

c. Use the fourth Taylor polynomial to bound the error.

For a polynomial of degree $n = 4$, the error $R_5(x)$ is defined by the formula $R_5(x) = \frac{f^{(5)}(c)}{5!}(x - a)^5$, where c exists somewhere in the interval $[0, \pi]$. We don't know the value of c , so we find an upper bound on $|R_5(0)|$ by determining the maximum value of $|f^{(5)}|$ on the interval $[0, \pi]$. Since $f^{(5)}(x) = \sin x$, the largest value for $|f^{(5)}(x)|$ on that interval is $\sin \frac{\pi}{2} = 1$. Therefore, we have $|R_5(0)| \leq \frac{1}{120}|\pi|^5 \approx 2.550164$

5.) If $fl(x)$ is the machine approximated number of a real number x and ϵ is the corresponding relative error, then show that $fl(x) = (1 - \epsilon)x$.

Let $fl(x)$ denote the machine approximation of a real number x . The relative error ϵ is defined as

$$\epsilon = \frac{x - fl(x)}{x}, \quad x \neq 0$$

Multiplying both sides by x gives

$$\epsilon x = x - fl(x)$$

Rearranging,

$$fl(x) = x - \epsilon x$$

Factoring out x , we obtain that

$$\boxed{fl(x) = x(1 - \epsilon)}$$

6.) For the following numbers x and their corresponding approximations x_A , find the number of significant digits in x_A with respect to x and find the relative error.

Note:

- s is the largest integer s.t. $\beta^s \leq |x|$
- If x_A is an approximation to x , then we say that x_A approximates x to r significant β -digits if $|x - x_A| \leq \frac{1}{2}\beta^{s-r+1}$
- The relative error is defined as $E_r = \frac{|x - x_A|}{x}$

a. $x = 451.01$, $x_A = 451.023$

For $x = 451.01$, the approximate number $x_A = 451.023$ has four significant digits since $|x - x_A| = |451.01 - 451.023| = 0.013 \leq 0.05 = \frac{1}{2}10^{-1}$, since $10^2 = 100 \leq 451.01 < 10^3$, we have $s = 2$. Therefore, in this case $\boxed{r = 4}$, because $|x - x_A| \leq \frac{1}{2}\beta^{s-r+1} = \frac{1}{2}\beta^{2-4+1} = \frac{1}{2}10^{-1}$

The relative error is $\boxed{E_r(x_A) = \frac{0.013}{451.01} \approx 2.88 \times 10^{-5}}$.

b. $x = -0.04518$, $x_A = -0.045113$

For $x = -0.04518$, the approximate number $x_A = -0.045113$ has two significant digits since $|x - x_A| = |(-0.04518) - (-0.045113)| = 0.000067 \leq 0.0005 = \frac{1}{2}10^{-3}$, since $10^{-2} = 0.01 \leq -0.04518 < 10^{-1}$, we have $s = -2$. Therefore, in this case $\boxed{r = 2}$, because $|x - x_A| \leq \frac{1}{2}\beta^{s-r+1} = \frac{1}{2}\beta^{-2-2+1} = \frac{1}{2}10^{-3}$

The relative error is $\boxed{E_r(x_A) = \frac{0.000067}{0.04518} \approx 1.48 \times 10^{-3}}$.

c. $x = 23.4604$, $x_A = 23.4213$

For $x = 23.4604$, the approximate number $x_A = 23.4213$ has three significant digits since $|x - x_A| = |23.4604 - 23.4213| = 0.0391 \leq 0.05 = \frac{1}{2}10^{-1}$, since $10^1 = 10 \leq 23.4604 < 10^2$, we have $s = 1$. Therefore, in this case $\boxed{r = 3}$, because $|x - x_A| \leq \frac{1}{2}\beta^{s-r+1} = \frac{1}{2}\beta^{1-3+1} = \frac{1}{2}10^{-1}$

The relative error is $\boxed{E_r(x_A) = \frac{0.0391}{23.4604} \approx 1.67 \times 10^{-3}}$.

7.) Find the condition number for the following functions

a. $f(x) = 2x^2$

$$f'(x) = 4x$$

$$cn = \left| \frac{x(4x)}{2x^2} \right| = 2$$

$$cn = 2$$

b. $f(x) = 2\pi^x$

$$f'(x) = 2\pi^x \ln(\pi)$$

$$cn = \left| \frac{x(2\pi^x \ln(\pi))}{2\pi^x} \right| = x \ln(\pi)$$

$$cn = x \ln(\pi)$$

c. $f(x) = 2b^x$

$$f'(x) = 2b^x \ln(b)$$

$$cn = \left| \frac{x(2b^x \ln(b))}{2b^x} \right| = x \ln(b)$$

$$cn = x \ln(b)$$

8.) Determine if the following series converges or diverges. If it converges determine its sum.

$$\sum_{i=1}^{\infty} \frac{1}{2^n}$$

This is a geometric series with a common ratio

$$r = \frac{1}{2}$$

Since $|r| < 1$, the series converge. The sum of geometric series starting at $n = 1$ is

$$\sum_{i=1}^{\infty} ar^n = \frac{ar}{1-r}$$

$$\sum_{i=1}^{\infty} ar^n = \frac{1/2}{1-1/2} = 1$$